PRODUCT AND QUOTIENT RULES

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Name: Solutions

In each of the problems, use the derivative rules to differentiate the given function. Use proper notation and simplify your final answers. In some cases it might be advantageous to simplify/rewrite first. Do not use rules found in later sections.

1. $f(x) = \left(1 + \sqrt{x}\right) x^3$

$$f'(x) = \frac{d}{dx}(1+x^{1/2})x^3 + (1+x^{1/2})\frac{d}{dx}x^3$$
$$= (0 + \tfrac{1}{2}x^{-1/2})x^3 + (1+x^{1/2})3x^2$$
$$= \boxed{\tfrac{1}{2}x^{5/2} + 3x^2 + x^{7/2}}$$

2. $g(t) = \left(\frac{2}{t} + t^5\right)(t^3 + 1)$

$$g'(t) = \frac{d}{dt}(2t^{-1} + t^5)(t^3+1) + \left(\tfrac{2}{t} + t^5\right)\frac{d}{dt}(t^3+1)$$
$$= (2(-1)t^{-2} + 5t^4)(t^3+1) + \left(\tfrac{2}{t} + t^5\right)(3t^2 + 0)$$
$$= \left(-\tfrac{2}{t^2} + 5t^4\right)(t^3+1) + \left(\tfrac{2}{t} + t^5\right)(3t^2)$$
$$= -2t - 2/t^2 + 5t^7 + 5t^4 + 6t + 3t^7$$
$$= \boxed{8t^7 + 5t^4 + 4t - 2/t^2}$$

3. $h(y) = \dfrac{1}{y^3 + 2y + 1}$

$$h'(y) = \frac{\frac{d}{dy}(1)(y^3+2y+1) - (1)\frac{d}{dy}(y^3+2y+1)}{(y^3+2y+1)^2}$$
$$= \frac{0 - (3y^2+2y)}{(y^3+2y+1)^2}$$
$$= \boxed{\frac{-3y^2-2y}{(y^3+2y+1)^2}}$$

4. $y = \dfrac{1}{x + \sqrt{x}}$

$$f(x) = 1, \quad f'(x) = 0, \quad g(x) = x + \sqrt{x} = x + x^{1/2}, \quad g'(x) = 1 + \tfrac{1}{2}x^{-1/2} = 1 + \frac{1}{2\sqrt{x}}$$
$$\frac{dy}{dx} = \frac{d}{dx}\frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} = \frac{0(x+\sqrt{x}) - (1)(1+\frac{1}{2\sqrt{x}})}{(x+\sqrt{x})^2}\frac{2\sqrt{x}}{2\sqrt{x}}$$
$$= \boxed{\frac{-2\sqrt{x} - 1}{2\sqrt{x}(x+\sqrt{x})^2}}$$

5. $g(z) = \dfrac{z^2 + 1}{z^3 - 5}$

$$g'(z) = \frac{\frac{d}{dz}(z^2+1)(z^3-5) - (z^2+1)\frac{d}{dz}(z^3-5)}{(z^3-5)^2}$$
$$= \frac{(2z+0)(z^3-5) - (z^2+1)(3z^2-0)}{(z^3-5)^2}$$
$$= \frac{2z^4 - 10z - 3z^4 - 3z^2}{(z^3-5)^2} = \boxed{\frac{-z^4 - 3z^2 - 10z}{(z^3-5)^2}}$$

6. $y = \dfrac{\sqrt{x}}{x^3 + 1}$

$$\frac{dy}{dx} = \frac{\frac{d}{dx}(x^{1/2})(x^3+1) - \sqrt{x}\frac{d}{dx}(x^3+1)}{(x^3+1)^2}$$
$$= \frac{(\frac{1}{2}x^{-1/2})(x^3+1) - \sqrt{x}(3x^2)}{(x^3+1)^2}$$
$$= \frac{\frac{1}{2}x^{5/2} + \frac{1}{2}x^{-1/2} - 3x^{5/2}}{(x^3+1)^2} = \frac{-\frac{5}{2}x^{5/2} + \frac{1}{2}x^{-1/2}}{(x^3+1)^2}\left(\frac{2\sqrt{x}}{2\sqrt{x}}\right)$$
$$= \boxed{\frac{-5x^3 + 1}{2\sqrt{x}(x^3+1)^2}}$$

Date: September 17, 2018.

$(t-4)(2-t^3) = 2t - t^4 - 8 + 4t^3$ $\quad \cdot t^2(-3t^3 + 12t^2)$

7. $z = \dfrac{t^2}{(t-4)(2-t^3)}$

$$\frac{dz}{dt} = \frac{2t(t-4)(2-t^3) - t^2(2-t^3 + (t-4)(-3t^2))}{(t-4)^2(2-t^3)^2}$$

$$= \frac{4t^2 - 2t^5 - 16t + 8t^4 - 2t^2 + t^5 + 3t^5 - 12t^4}{(t-4)^2(2-t^3)^2}$$

$$= \frac{(-2+1+3)t^5 + (8-12)t^4 + (4-2)t^2 - 16t}{(t-4)^2(2-t^3)^2} = \boxed{\frac{2t^5 - 4t^4 + 2t^2 - 16t}{(t-4)^2(2-t^3)^2}}$$

8. $h(x) = \dfrac{(x^3+1)\sqrt{x}}{x^2}$

$$h(x) = \frac{(x^3+1)x^{1/2}}{x^2} = \frac{x^{7/2} + x^{1/2}}{x^{4/2}} = \frac{x^{7/2}}{x^{4/2}} + \frac{x^{1/2}}{x^{4/2}}$$

$$= x^{7/2 - 4/2} + x^{1/2 - 4/2} = x^{3/2} + x^{-3/2}$$

$$h'(x) = 3/2\, x^{3/2 - 2/2} + (-3/2)x^{-3/2 - 2/2}$$

$$= \boxed{3/2\, x^{1/2} - 3/2\, x^{-5/2}}$$

9. Let $f(x) = g(x)h(x)$, $g(10) = -4$, $h(10) = 560$, $g'(10) = 0$, $h'(10) = 4$. Find $f'(10)$.

$$f'(10) = g'(10)h(10) + g(10)h'(10)$$

$$= 0(560) + (-4)(4) = \boxed{-16.}$$

10. Let $z(-3) = 6$, $z'(-3) = 15$, and

$$y(x) = \frac{z(x)}{1+x^2}.$$

Find $y'(-3)$.

$$y'(x) = \frac{z'(x)(1+x^2) - z(x)(2x)}{(1+x^2)^2}$$

$$y'(-3) = \frac{z'(3)(1+9) - z(3)(-6)}{(1+9)^2} = \frac{15(10) + 6(6)}{10^2} = \frac{150+36}{100} = \boxed{\frac{186}{100} = 1.86.}$$